

“The future seems to be in how the devices integrate themselves in a hospital ecosystem and their ability to communicate with other systems—Rohit Mehta, general manager, medical equipment & systems, Larsen & Toubro



ficient Bluetooth solutions for medical device manufacturers. Device manufacturers integrate these solutions into their products and make them connect through smart phones.

MindTree's Cardius is a suite of advanced software tools for objective evaluation of cardiac functionality. The assessed results aid clinicians in detecting, diagnosing and managing cardiac diseases. Cardius performs the assessment by non-invasively measuring specific parameters of the chambers of the heart.

Texas Instruments. Texas Instruments offers analogue front ends (for ultrasound and digital X-ray equipment), single-/multi-core processors for real-time image processing (for ultrasound, MRI and CT/PET systems), low-power microcontrollers for portable/consumer medical electronics (digital thermometers, ECG and pulse-oximeters) and low-power, analogue front end for bipotential measurements (ECG and EEG).

Freescall. Freescale leverages its expertise in microcontrollers, sensors, and analogue and wireless products to offer a wide array of embedded technologies ideal for use in home portable medical applications such as diagnostic and therapy devices and medical imaging devices.

Freescall provides solutions for products including portable home devices (blood pressure monitors, fetal heart rate monitors, heart rate monitors, pulse oximetry, blood glucose monitors, telehealth system, hearing aids and portable ECG), diagnostics and therapy (ECG, ventilators and respirators, defibrillators, vital signs monitors and anaesthesia unit monitor), imaging (ultrasound and digital X-ray) and intelligent hospitals (hospita-

tal admission machines and powered patient beds).

Freescale design resources include software and tools (CodeWarrior development tools, IEEE 802.15.4 (Zig-Bee) protocol stacks and tools, medical applications USB stack, sensor toolbox and touch sensing software suite) and technologies, standards and protocols (IEEE 802.15.4 protocol, acceleration sensors, pressure sensors and proximity sensors).

Magma Design Automation. Magma's EDA software enables semiconductor companies to design chips with low power consumption in minimal area and time.

Majority of medical electronic devices broadly contain three categories of semiconductor chips: chips that capture the input data and convert it for processing, digital processing chips that process the data, and imaging chips for display of diagnostic results and video during complex surgical procedures.

Magma's Talus family of tools is used extensively in the design of digital processing, wireless modem and imaging chips that form the core of the medical electronic equipment. The Titan family of tools enables the design of converters and amplifiers that capture and convert the input data.

Xilinx. Xilinx programmable chips enable medical equipment designers to address the escalating system perfor-

mance requirements in next-generation imaging systems. The Virtex FPGA family delivers the right mix of flexibility, hard intellectual property (IP) cores, transceiver capabilities and development tool support to help them meet market demands with evolving standards and stringent performance requirements.

For diagnostic, monitoring and therapy applications, the Virtex FPGA and Spartan FPGA families can be used to meet a range of processing, display and I/O interface requirements. For home medical devices, Spartan FPGAs offer a balance of low cost, low power and high integration. Xilinx Imaging Pipe IP for medical imaging applications supports up to 1080p HD at 60 fps with a modular structure that allows for customisation or addition. The Xylon logiTAP platform implements high-quality, low-cost graphical user interfaces in embedded systems using low-cost Xilinx Spartan-3E FPGAs.

STMicroelectronics. STMicroelectronics is addressing medical field with the following key areas where electronics can be used: automated external defibrillators, blood analysers, blood pressure meters, cholesterol meters, computed tomography imaging, disposable DNA pathogen detectors, glucose meters, hearing aids, infusion pumps, implantable cardioverter defibrillators, magnetic resonance imaging, pacemakers, patient monitoring, positron emission tomography imaging (PET), thermometers and ultrasound imaging.

For the portable healthcare market, the company has a broad portfolio of devices such as analogue and power ICs, MEMS and low-power micro-

“Semiconductors will play a critical role in the development of new technologies that assist in patient monitoring, diagnostics, therapy and imaging—Vivek Tyagi, country sales manager, Freescale Semiconductor India

